

Chemistry Chapter 10

Acids, Bases aur Salts

Aasan Notes, tumhari kitab ke saath saath (book pages 140 to 149)

Assalam o alaikum Seemab! Ye chapter class mein ho chuka hai. Ye notes usay **bilkul pakka** karne ke liye hain, bohat aasan alfaaz aur ghar ki misalon ke saath. Headings aur page numbers kitab wale hi hain.

Pichle paper mein jo 3, 4 cheezein thori mushkil lagi thin, un par khaas  **"Pakki trick"** boxes hain. Unhein zaroor parhna!

Kaise parhna hai (sirf 30 se 35 minute):

1. Ek dafa aaram se parho. Har hisse ke aakhir mein **"Ek line mein"** box hai: bas woh yaad rahe to kaafi hai.
2. Phir neelay  boxes dobara parho: ye kitab ke English alfaaz hain, exam mein yehi likhne hain.
3. Aakhir mein 12 sawal khud karo, jawab agle page par hain.

Is chapter ka naqsha (6 hisse)

 Hissa 1	Acid aur base pehchano (10.1, Table 10.1)
 Hissa 2	Arrhenius concept: paani mein H^+ ya OH^- (10.1.1)
 Hissa 3	Bronsted-Lowry concept: proton dene wala, lene wala (10.1.2)
 Hissa 4	Strong aur weak; alkalis (10.2)
 Hissa 5	Acids aur bases ke kaam (reactions) (10.3, 10.4)
 Hissa 6	Oxides aur hydroxides (10.5)



Hissa 1: Acid aur base pehchano

Introduction (kitab p141)

Acids aur bases hamare ghar mein har jagah hain:

Acids wali cheezein (khatti)	Bases wali cheezein
Vinegar (sirka), lemon juice (nimbu), tomato, apple, cola drinks, aspirin, toilet cleaners	Drain cleaner, antacid tablets, baking powder, soda

Hamara jism bhi acid banata hai (pet mein). "Acid indigestion" (tezabiyat) aur "acid rain" jaise alfaaz tum ne sunay honge.

10.1 Acids aur bases ki pehchan (kitab p141)

Table 10.1 (poori yaad karo, 5 lines hain bas):

Property	Acid	Base
Taste (zaiqa)	Sour (khatta)	Bitter (kadwa)
Blue litmus par	Turns red	No effect
Red litmus par	No effect	Turns blue
Skin (jild) par	Corrosive	Corrosive
Electrical conductivity	Aqueous solutions conduct electricity	Aqueous solutions conduct electricity



Ghar ki misal

Nimbu aur imli **khatte** hain: acids. Sabun ko zabaan par lagao (mat lagana!) to **kadwa** lagega aur haath mein **phisalta** hai: base.



Litmus ki trick

Base aur **Blue**, dono **B** se! Base red litmus ko **blue** karta hai.
Acid bilkul ulta: blue litmus ko **red** karta hai.

🌟 Ek line mein

Acid = khatta, blue ko red karta hai. Base = kadwa, red ko blue karta hai.
Dono corrosive, dono ke solutions bijli guzarte hain.

💧 Hissa 2: Arrhenius concept

10.1.1 Arrhenius concept (kitab p141, 142)

1887 mein Sweden ke chemist **Svante Arrhenius** ne acids aur bases ki pehli kamyab theory di. Ye theory **paani** ke baare mein hai: cheez ko paani mein daalo aur dekho kya nikalta hai.

📖 **Kitab ki definition: exam mein yehi likho**

"An acid is a substance that ionizes in water to produce H^+ ions and a base is a substance that ionizes in water to produce OH^- ions."

🍪 **Ionize ka matlab: biscuit todna**

Ionize = molecule ka **tootna**, ions mein. Jaise biscuit todo: **poora biscuit pehle** (left side), **tukre baad mein** (right side).

🎯 **Pakki trick: equation ULTI mat likhna**

Pichle paper mein equation ulti likh di thi (" $H + NO_3 \rightarrow HNO_3$ "). Hamesha yaad rakho:

Poora molecule LEFT par → toote hue ions RIGHT par.

Sahi: $HNO_3 \rightarrow H^+ + NO_3^-$ Ghalat: $H^+ + NO_3^- \rightarrow HNO_3$

Kitab ki equations (arrow ke upar H_2O ka matlab: paani mein):

Cheez	Paani mein tootne ki equation
HCl (acid)	$\text{HCl(g)} \rightarrow \text{H}^+(\text{aq}) + \text{Cl}^-(\text{aq})$
HNO_3 (acid)	$\text{HNO}_3(\text{l}) \rightarrow \text{H}^+(\text{aq}) + \text{NO}_3^-(\text{aq})$
H_2SO_4 (acid)	$\text{H}_2\text{SO}_4(\text{l}) \rightarrow 2\text{H}^+(\text{aq}) + \text{SO}_4^{2-}(\text{aq})$
NaOH (base)	$\text{NaOH(s)} \rightarrow \text{Na}^+(\text{aq}) + \text{OH}^-(\text{aq})$
KOH (base)	$\text{KOH(s)} \rightarrow \text{K}^+(\text{aq}) + \text{OH}^-(\text{aq})$
NH_4OH (base)	$\text{NH}_4\text{OH(aq)} \rightleftharpoons \text{NH}_4^+(\text{aq}) + \text{OH}^-(\text{aq})$

H_2SO_4 mein 2 H hain, is liye 2H^+ nikalte hain. Acid se hamesha H^+ , base se hamesha OH^- .

Table 10.2 aur 10.3: common acids, bases aur un ke kaam

Naam	Formula	Kaam (use)	 Trick
Hydrochloric acid	HCl	Cleaning of metals, bricks and removing scale from boilers	Cl = Clean
Nitric acid	HNO_3	Fertilizers, explosives	N = Nitrogen khaad
Sulphuric acid	H_2SO_4	Many chemicals, drugs, dyes, paints and explosives	S = Sab kuch
Phosphoric acid	H_3PO_4	Fertilizers, acidulant for food	P = Plants aur Plate (khana)
Sodium hydroxide	NaOH	Soap making, drain cleaners	Na = Nahane ka sabun
Potassium hydroxide	KOH	Liquid soap, shaving cream	K = Kreem
Calcium hydroxide	Ca(OH)_2	Mortar, plasters, cement	Ca = Construction
Magnesium hydroxide	Mg(OH)_2	Antacid, laxative	Mg = Medicine

✨ Ek line mein

Arrhenius (1887): paani mein H^+ de = acid, paani mein OH^- de = base.
Molecule left par, ions right par.



Hissa 3: Bronsted-Lowry concept

10.1.2 Arrhenius theory ki kami (kitab p142)

Arrhenius theory sirf **paani wale solutions** (aqueous) par chalti hai. Aur ye nahin bata sakti ke:

- **CO_2 aur SO_2 acidic kyun hain?** Un mein to koi H hi nahin!
- **NH_3 base kyun hai?** Us mein to koi OH hi nahin!

Is kami ko door karne ke liye **1923** mein **J. N. Bronsted** aur **T. M. Lowry** ne alag alag (independently) nayi theory di.



Kitab ki definition: exam mein yehi likho

"According to this theory, an acid is a proton donor and a base is a proton acceptor."



Chocolate wali misal (sab se aasan)

Proton = H^+ = ek chocolate.

Jo chocolate **DETA** hai (donor) woh **ACID**.

Jo chocolate **LETA** hai (acceptor) woh **BASE**.

Bas itni si baat hai!

Pakki trick: H gino!

Pichle paper mein kabhi paani ka role ulta ho gaya, kabhi H_3O^+ ko acid likh diya. Ab ye 3 steps karo:

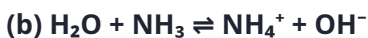
1. Sirf **LEFT side** wali cheezon ko acid ya base kaho. Right side (H_3O^+ , Cl^- , NH_4^+ , OH^-) ko kabhi nahin.
2. Left wali cheez ka right side par kya bana? **H kam hua** (chocolate de di) → woh **ACID**.
3. **H barh gaya** (chocolate mil gayi) → woh **BASE**.

Example 10.1 (kitab p143): trick laga kar dekho



$\text{HCl} \rightarrow \text{Cl}^-$ H kam hua, chocolate di **$\text{HCl} = \text{acid}$**

$\text{H}_2\text{O} \rightarrow \text{H}_3\text{O}^+$ H barha, chocolate li **$\text{H}_2\text{O} = \text{base}$**



$\text{H}_2\text{O} \rightarrow \text{OH}^-$ H kam hua, chocolate di **$\text{H}_2\text{O} = \text{acid}$**

$\text{NH}_3 \rightarrow \text{NH}_4^+$ H barha, chocolate li **$\text{NH}_3 = \text{base}$**

Dekha? **Paani (a) mein base tha aur (b) mein acid!** Isi liye kitab kehti hai: water behaves like an acid as well as a base. It is **amphoteric** in nature.

Kitab ki definition

“Substances that react with both acids and bases are called amphoteric substances.”

Paani ki misal

Paani us saheli jaisa hai jo HCl ke saath ho to chocolate **le leti** hai (base), aur NH_3 ke saath ho to chocolate **de deti** hai (acid). Saathi dekh kar role badalta hai!

★ Kitab ki problem solving strategy (p143)

1. Acid proton de kar **negative ion** ban jata hai ($\text{HCl} \rightarrow \text{Cl}^-$).
 2. Base proton le kar **positive ion** ban jata hai ($\text{NH}_3 \rightarrow \text{NH}_4^+$, $\text{H}_2\text{O} \rightarrow \text{H}_3\text{O}^+$).
- Aur yaad rakho: Arrhenius wale **sab acids** Bronsted-Lowry mein bhi acids hain.

💪 Concept Assessment Exercise 10.1 (kitab p144): jawab ke saath

1. $\text{H}_2\text{SO}_4 + \text{H}_2\text{O} \rightarrow \text{HSO}_4^- + \text{H}_3\text{O}^+$ **H_2SO_4 acid, H_2O base**
2. $\text{CH}_3\text{COOH} + \text{H}_2\text{O} \rightleftharpoons \text{CH}_3\text{COO}^- + \text{H}_3\text{O}^+$ **CH_3COOH acid, H_2O base**
3. $\text{H}_2\text{S} + \text{NH}_3 \rightleftharpoons \text{NH}_4^+ + \text{HS}^-$ **H_2S acid, NH_3 base**

Har ek mein trick lagao: kis ka H kam hua, kis ka barha?

✨ Ek line mein

Bronsted-Lowry (1923): acid = **proton donor** (H deta hai), base = **proton acceptor** (H leta hai). Paani dono ban sakta hai: **amphoteric**.

💪 Hissa 4: Strong, weak aur alkalis

10.2 Strength of acids and bases (kitab p144)

Sab acids aur bases paani mein **ek jitna nahin toot-te**. Is liye kuch strong hain, kuch weak.

📖 Kitab ki definitions

"An acid that ionizes completely in aqueous solution is called a strong acid."

"Acids that do not ionize completely in aqueous solutions are called weak acids."

"A base that ionizes completely or 100% in aqueous solution is termed as a strong base."

"A base that ionizes to a little extent is called a weak base."

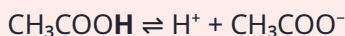
	Strong (100% toot-ta)	Weak (thora sa toot-ta)
Acids	HCl, HNO ₃ , H ₂ SO ₄	CH ₃ COOH (ethanoic / acetic acid, sirke mein): sirf 5% tak
Bases	NaOH, KOH, Ca(OH) ₂	Al(OH) ₃ , NH ₃ (kam OH ⁻ dete hain)
Arrow	Ek arrow →	Do arrow ⇌

Biscuit packet ki misal

Strong acid = poore packet ke **sab** biscuit toot gaye (100%).

Weak acid = 100 biscuits mein se sirf **5** toote (sirke wala acetic acid).

Pakki trick: CH₃COOH ka sirf AAKHRI H alag hota hai



CH₃ wale H kabhi alag nahin hote. Toota hua hissa poora likho: **CH₃COO⁻** (CH₃ mat bhoolna). Aur weak hai, is liye ⇌ lagao.

Alkalis (kitab p145)

Kitab ki definition

“A base which is soluble in water is called an alkali.”

- **Sab alkalis bases hain, lekin sab bases alkalis nahin.**
- Paani mein na ghulne wale bases: **Cu(OH)₂** (copper hydroxide), **Al(OH)₃** (aluminium hydroxide), **Fe(OH)₃** (ferric hydroxide).
- Ghar ke alkalis: **soaps, detergents, shampoos, toothpaste.**
- KOH ya NaOH, kaun alkali? **Dono**, kyunke dono paani mein ghul jate hain.

Aasan misal

Har Lahori Pakistani hai, lekin har Pakistani Lahori nahin. Isi tarah har **alkali** base hai, lekin har base alkali nahin!

✨ Ek line mein

Strong = poora toot-ta ($\rightarrow$), weak = thora sa ($\rightleftharpoons$). Paani mein ghulne wala base = **alkali**.



Hissa 5: Acids aur bases ke kaam

10.3 Characteristic properties of acids (kitab p145, 146)

Acids ki 5 pehchan:

1. Acids have **sour taste**.
2. Acids change the colour of **blue litmus paper to red**.
3. **Metals ke saath:** salt + **hydrogen gas** (bulbule). Zinc, magnesium, aluminium, calcium jaise metals.
4. **Metal carbonates ke saath:** salt + water + **carbon dioxide** (bulbule).
5. **Bases ke saath:** salt + water. Isay **neutralization** kehte hain.

🧠 Teen reactions, ek trick: "Salt har jagah!"

Acid + **Metal** $\rightarrow$ Salt + **Hydrogen**

Acid + **Carbonate** $\rightarrow$ Salt + Water + **CO₂** (Carbonate se CO₂)

Acid + **Base** $\rightarrow$ Salt + **Water** (neutralization)

🎯 Pakki trick: "2" lagana mat bhoolna (kitab bhi bhool gayi!)

Kitab ne 3 misalon mein balance karna chhor diya hai. **Tum sahi wali likhna:**

Kitab mein chhapa (ghalat) ✗	Sahi, balanced ✓
$\text{Na}_2\text{CO}_3 + \text{HCl} \rightarrow \text{NaCl} + \text{H}_2\text{O} + \text{CO}_2$	$\text{Na}_2\text{CO}_3 + 2\text{HCl} \rightarrow 2\text{NaCl} + \text{H}_2\text{O} + \text{CO}_2$
$\text{H}_2\text{SO}_4 + \text{KOH} \rightarrow \text{K}_2\text{SO}_4 + \text{H}_2\text{O}$	$\text{H}_2\text{SO}_4 + 2\text{KOH} \rightarrow \text{K}_2\text{SO}_4 + 2\text{H}_2\text{O}$
$\text{H}_2\text{SO}_4 + \text{NaOH} \rightarrow \text{Na}_2\text{SO}_4 + \text{H}_2\text{O}$	$\text{H}_2\text{SO}_4 + 2\text{NaOH} \rightarrow \text{Na}_2\text{SO}_4 + 2\text{H}_2\text{O}$



Balance kaise karein: atoms gino

$\text{Na}_2\text{CO}_3 + \text{HCl} \rightarrow \text{NaCl} + \text{H}_2\text{O} + \text{CO}_2$ (abhi balanced nahin)

Na: left 2, right 1 → right par **2NaCl**

Cl: ab right 2, left 1 → left par **2HCl**

H: left 2, right 2 **barabar**

C, O: C 1 = 1, O 3 = 3 **barabar**

$\text{Na}_2\text{CO}_3 + 2\text{HCl} \rightarrow 2\text{NaCl} + \text{H}_2\text{O} + \text{CO}_2$ Done!

Aakhir mein har atom ek ek kar ke gin lo. 30 second lagte hain aur poore marks milte hain.

Kitab ki baqi misalein (ye pehle se balanced hain):

Reaction	Equation
Acid + metal	$\text{Zn} + 2\text{HCl} \rightarrow \text{ZnCl}_2 + \text{H}_2$
Acid + metal	$\text{Mg} + \text{H}_2\text{SO}_4 \rightarrow \text{MgSO}_4 + \text{H}_2$
Acid + carbonate	$\text{CaCO}_3 + \text{H}_2\text{SO}_4 \rightarrow \text{CaSO}_4 + \text{H}_2\text{O} + \text{CO}_2$ CaCO_3 = limestone ya marble. Ye reaction glass, paper aur soap banane mein kaam aati hai.
Neutralization	$\text{HCl} + \text{NaOH} \rightarrow \text{NaCl} + \text{H}_2\text{O}$



Neutralization ki ghar wali misal

Pet mein tezabiyat (acidity)? Doctor **antacid** deta hai. Antacid mein base hota hai, jaise **$\text{Mg}(\text{OH})_2$** (milk of magnesia). Ye pet ke acid (HCl) ko **neutralize** kar deta hai: acid + base → salt + water. Aag par paani jaisa!

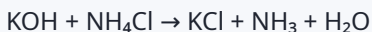
Acid rain (kitab p145, 146):

- **Normal baarish** bhi thori si acidic hoti hai, kyunke us mein **carbon dioxide** ghuli hoti hai.
- Insaan ki activities se **acidic oxides** hawa mein jaate hain aur baarish ka paani **zyada acidic** ho jata hai: ye **acid rain** hai.
- Acid rain mein **nitric acid** aur **sulphuric acid** ghule hote hain.
- Acids **corrosive** hain, is liye acid rain **metals aur metal carbonates** wali buildings, structures aur statues ko nuqsan deti hai. (Is ki poori kahani

10.4 Characteristic properties of bases (kitab p146)

1. Bases have **bitter taste**.
2. Bases change the colour of **red litmus paper to blue**.
3. Aqueous solution of bases have **slippery touch**. (Sabun wala paani haath par phisalta hai!)
4. Bases **neutralize acids** to form salt and water: $\text{H}_2\text{SO}_4 + 2\text{NaOH} \rightarrow \text{Na}_2\text{SO}_4 + 2\text{H}_2\text{O}$
5. Bases **ammonium salts** ko garam karne par tod dete hain aur **ammonia gas** nikalti hai.

Base + Ammonium salt $\rightarrow$ Salt + Ammonia + Water



🔗 **Pakki trick: gas ka NAAM bhi likho**

Sirf " NH_3 " likhna kaafi nahin. Jab sawal gas poochhe to alag se likho:

"The gas released is ammonia (NH_3)."

Yaad kaise rakhein? **Ammonium** salt se **Ammonia** gas. Naam mein hi likha hai!

🌟 **Ek line mein**

Acid + metal = H_2 ; acid + carbonate = CO_2 ; acid + base = salt + water (neutralization). Base + ammonium salt = ammonia gas. Balance karna aur gas ka naam, dono mat bhoolo.



Hissa 6: Oxides aur hydroxides

10.5 Oxides and hydroxides (kitab p146)

Bases = metals ke oxides aur hydroxides. Metal ko oxygen mein jalao to **metal oxide** banta hai.

Metal + Oxygen → Metal oxide

Equation	Naam (English mein bhi likho!)
$4\text{Na} + \text{O}_2 \rightarrow 2\text{Na}_2\text{O}$	Na_2O = sodium oxide
$2\text{Mg} + \text{O}_2 \rightarrow 2\text{MgO}$	MgO = magnesium oxide

🔗 Pakki trick: oxygen hamesha JORI mein

Oxygen gas hamesha **O₂** likhi jati hai, jaise jutay hamesha jori mein. Akeli "O" kabhi nahin.

Naam ka formula: **metal ka naam + "oxide"**. Na_2O = sodium oxide, MgO = magnesium oxide, CaO = calcium oxide.

Metal oxide + paani → metal hydroxide. Oxide ion paani se proton (H^+) le leta hai aur hydroxide ion ban jata hai. Yani oxide yahan **Bronsted base** hai aur paani **proton donor** (acid).



Equation	Naam
$\text{MgO} + \text{H}_2\text{O} \rightarrow \text{Mg(OH)}_2$	magnesium hydroxide
$\text{Na}_2\text{O} + \text{H}_2\text{O} \rightarrow 2\text{NaOH}$	sodium hydroxide
$\text{CaO} + \text{H}_2\text{O} \rightarrow \text{Ca(OH)}_2$	calcium hydroxide

- **Zyada tar** metal oxides aur hydroxides **basic** hote hain.
- Lekin kuch paani mein nahin ghulte aur **acid aur base dono** jaisa kaam karte hain: **amphoteric oxides and hydroxides**.
- Misalein: **Al_2O_3** (aluminium oxide), **Al(OH)_3** , **ZnO** (zinc oxide), **Zn(OH)_2** .

Amphoteric ki trick: "A se Z tak"

Aluminium aur **Z**inc: A se Z, yani **dono taraf!** In ke oxides aur hydroxides acid aur base dono ke saath react karte hain.

Ek line mein

Metal + $O_2 \rightarrow$ metal oxide; metal oxide + paani $\rightarrow$ metal hydroxide (base).
Aluminium aur zinc wale = amphoteric.



Key Points: kitab ke alfaaz mein

- According to Arrhenius theory, an acid is a substance that ionizes in water to produce H^+ ions and a base is a substance that ionizes in water to produce OH^- ions.
- A Bronsted-Lowry acid is a proton donor and a base is a proton acceptor.
- A strong acid completely dissociates in aqueous solution and a weak acid dissociates partially in aqueous solution.
- A strong base completely dissociates in aqueous solution and a weak base dissociates partially in aqueous solution.
- Bases are oxides and hydroxides of metals.
- Alkalis are water soluble bases.
- Acids neutralize bases to form salt and water.
- Acids decompose carbonates to form salt, water, and carbon dioxide.
- Acids corrode metals and form salt and hydrogen.

Kitab ke Review Questions mein 2 ghaltiyan (p148, Q4)

(ii) "Sulphuric acid + Sodium carbonate $\rightarrow$ Sodium sulphate + water":
carbon dioxide bhi banti hai. Sahi: $H_2SO_4 + Na_2CO_3 \rightarrow Na_2SO_4 + H_2O + CO_2$

(iii) "Hydrochloric acid + Magnesium $\rightarrow$ Magnesium chloride + water":
water nahin, **hydrogen** banti hai. Sahi: $Mg + 2HCl \rightarrow MgCl_2 + H_2$



Poora chapter, ek page mein

Acid	Khatta, blue litmus $\rightarrow$ red, H^+ deta hai (Arrhenius), proton deta hai (Bronsted)
Base	Kadwa, phisalne wala, red litmus $\rightarrow$ blue, OH^- deta hai (Arrhenius), proton leta hai (Bronsted)
Amphoteric	Acid aur base dono: paani; Al aur Zn ke oxides/hydroxides
Strong	100% ionize ($\rightarrow$): HCl, HNO_3 , H_2SO_4 ; NaOH, KOH, $Ca(OH)_2$
Weak	Thora sa ionize ($\rightleftharpoons$): CH_3COOH (5%); $Al(OH)_3$, NH_3
Alkali	Paani mein ghulne wala base (soap, shampoo, toothpaste)
Acid + metal	Salt + H_2
Acid + carbonate	Salt + H_2O + CO_2
Acid + base	Salt + H_2O (neutralization)
Base + NH_4Cl	Salt + NH_3 (ammonia) + H_2O
Metal + O_2	Metal oxide; + $H_2O \rightarrow$ metal hydroxide
Acid rain	Nitric + sulphuric acid wali baarish; metals aur carbonates ko khaati hai

Paper se pehle 4 pakki baatein

1. Ionization: molecule **left**, ions **right**.
2. Bronsted: **H gino**; sirf left wali cheezon ko acid/base kaho.
3. Equation ke baad **atoms gino**, "2" lagao; oxygen hamesha **O_2** .
4. Gas ka aur banne wali cheez ka **naam English mein** bhi likho.

✓ Khud ko check karo (12 sawal)

Pehle notebook mein khud jawab likho, phir agle page par check karo.

1. Base red litmus ko kis rang ka karta hai?
2. Arrhenius ke mutabiq acid paani mein kaunsa ion deta hai?
3. HCl ke paani mein ionize hone ki equation likho (state symbols ke saath).
4. Bronsted-Lowry acid ki definition likho.
5. $\text{NH}_3 + \text{H}_2\text{O} \rightleftharpoons \text{NH}_4^+ + \text{OH}^-$ mein Bronsted acid aur base batao.
6. $\text{HNO}_3 + \text{H}_2\text{O} \rightarrow \text{H}_3\text{O}^+ + \text{NO}_3^-$ mein kya H_3O^+ acid hai? (Soch kar!)
7. Sirke mein kaunsa acid hai, aur woh strong hai ya weak?
8. Alkali kya hota hai? Ek aisa base batao jo alkali nahin.
9. Balance karo: $\text{Na}_2\text{CO}_3 + \text{HCl} \rightarrow \text{NaCl} + \text{H}_2\text{O} + \text{CO}_2$
10. $\text{NaOH} + \text{NH}_4\text{Cl}$ ko garam karo to kaunsi gas nikalti hai?
11. $2\text{Mg} + \text{O}_2 \rightarrow 2\text{MgO}$: MgO ka English naam kya hai?
12. Do amphoteric oxides ke naam batao.

👤 Agar kuch samajh na aaye (optional)

NBF Chemistry 9 Chapter 10 exercise ki video: youtube.com/watch?v=zELXCuDYFWI

Notes kaafi hain; video sirf tab dekho jab koi hissa mushkil lage.

Jawab

1. **Blue** (B se Base, B se Blue)
2. **H⁺ ion**
3. **HCl(g) → H⁺(aq) + Cl⁻(aq)** (molecule left, ions right)
4. **An acid is a proton donor.**
5. $\text{NH}_3 \rightarrow \text{NH}_4^+$ (H barha): **NH₃ = base**. $\text{H}_2\text{O} \rightarrow \text{OH}^-$ (H kam hua): **H₂O = acid**.
6. **Nahin!** H_3O^+ right side par hai. Acid **HNO₃** hai (H kam hua), base **H₂O** hai (H barha).
7. **Ethanoic acid (acetic acid), CH₃COOH**. Ye **weak** hai, sirf 5% tak ionize hota hai.
8. Paani mein ghulne wala base. Jo alkali nahin: **Cu(OH)₂, Al(OH)₃ ya Fe(OH)₃** (koi ek).
9. **Na₂CO₃ + 2HCl → 2NaCl + H₂O + CO₂**
10. **Ammonia gas (NH₃)**
11. **Magnesium oxide**
12. **Aluminium oxide (Al₂O₃) aur zinc oxide (ZnO)**

Score dekho:

- **10 se 12 sahi:** zabardast! 🌟 Chapter 10 tumhara pakka ho gaya.
- **7 se 9 sahi:** bohat achha! Jo ghalat hue, sirf un ka "Pakki trick" box dobara dekh lo.
- **7 se kam:** koi baat nahin. Kal ek dafa aur parho, doosri dafa hamesha aasan lagta hai. 💪